MIL-C-11272/9B
7 January 1970
SUPERSEDING
MIL-C-11272/9A
23 February 1961

MILITARY SPECIFICATION

CAPACITORS, FIXED, GLASS DIELECTRIC,

STYLE CY21

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION. USE MIL-C-23269.

1. SCOPE

1.1 This specification covers the detail requirements for style CY21, axial-radial lead, glass-dielectric (vitreous enamel), fixed capacitors.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-C-11272 Capacitors, Fixed, Glass Dielectric, General Specification for.

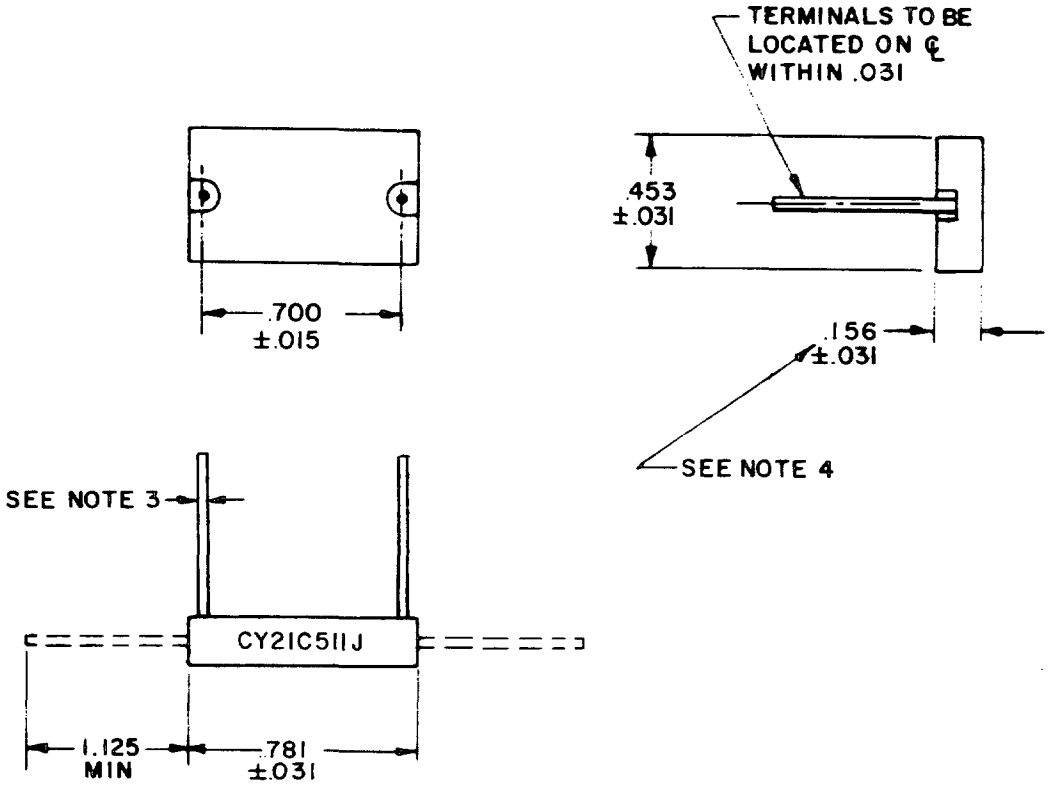
(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Requirements.** Requirements shall be in accordance with MIL-C-11272, and as specified herein.

3.2 **Design and construction.** Capacitors shall be of the design, construction, and physical dimensions specified on figure and in table I.

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.031	.79	.156	3.96	.700	17.78
.004	.10	.032	.81	.219	5.56	.781	19.84
.015	.38	.037	.94	.453	11.51	1.125	28.58

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. The lead diameter shall be $.032 \pm^{.004}_{.001}$ for capacitance values below above 1600 pF.
4. For any value above 1600 pF, this dimension shall be .219 ±.031.

Style CY21 capacitors.

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TABLE I. *Style CY21.*

Type designation 1/	Capacitance	DC rated voltage	Capacitance tolerances available
	pF	*volts*	*(±)*
CY21C511-	510	500	FGJ
CY21C561-	560	500	FGJ
CY21C621-	620	500	FGJ
CY21C681-	680	500	FGJ
CY21C751-	750	500	FGJ
CY21C821-	820	500	FGJ
CY21C911-	910	500	FGJ
CY21C102-	1,000	500	FGJ
CY21C112-	1,100	300	FGJ
CY21C122-	1,200	300	FGJ
CY21C132-	1,300	300	FGJ
CY21C152-	1,500	300	FGJ
CY21C162-	1,600	300	FGJ
CY21C182-	1,800	300	FGJ
CY21C202-	2,000	300	FGJ
CY21C222-	2,200	300	FGJ
CY21C242-	2,400	300	FGJ

1/ Complete type designation will include an additional letter symbol to indicate capacitance tolerance.

3.3 Life.

3.3.1 *For qualification inspection.* Capacitors shall meet the following requirements, as applicable:

(a) At 25°C:
  (1) Insulation resistance – 100,000 megohms or 100 ohmfarads, whichever is less.
  (2) Dissipation factor – Shall not exceed 0.002.
(b) At 125°C:
  (1) Insulation resistance – 10,000 megohms or 10 ohmfarads, whichever is less.
  (2) Dissipation factor – Shall not exceed 0.003.

3.3.2 *For quality conformance inspection.*

3.3.2.1 *Performance check.* Capacitors shall meet the following requirements:

(a) Insulation resistance – 100,000 megohms or 100 ohmfarads, whichever is less.
(b) Dissipation factor – Shall not exceed 0.002.

3.4 Quality factor (Q). The Q shall be not less than 2,000.

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3.5 Temperature coefficient and capacitance drift. The temperature coefficient and capacitance drift shall be within the limits specified in table II.

TABLE II. *Temperature coefficient and capacitance drift.*

Temperature coefficient	Capacitance drift (-55° to +125°C)
Parts/million/°C 105 ± 25	0.1 percent or 0.1 pF, whichever is greater.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-C-11272.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-C-11272.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-C-11272 are applicable to this specification.

Custodians:
Army – EL
Navy – EC
Air Force – 11

Review activities:
Army – EL
Navy – EC, SH
Air Force – 11, 17, 85
DSA – ES

User activities:
Army –
Navy –
Air Force – 19

Preparing activity:
Air Force – 11

Agent:
DSA – ES

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